

.NET 10 for distributed cloud and enterprise applications

Marc Müller
Principal Consultant

marc.mueller@4tecture.ch
@muellermarc
www.4tecture.ch

MVP Microsoft Most Valuable Professional

4tecture
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About me:

Marc Müller
Principal Consultant
@muellermarc

MVP
Microsoft Most Valuable Professional

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Our Products:

Multi-Tenant OpenID Connect Identity Provider

ProAuth

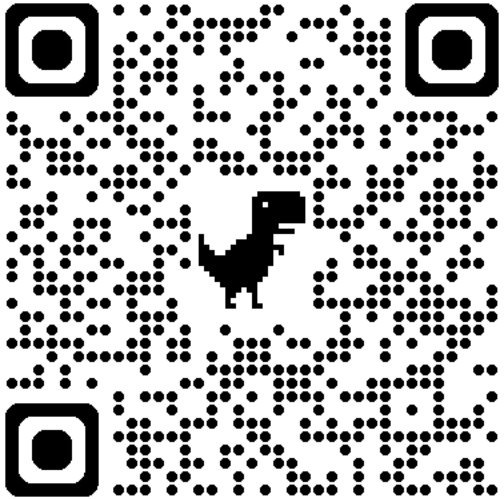
www.proauth.net

Enterprise Application Framework for .NET

ReaFx

www.reafx.net

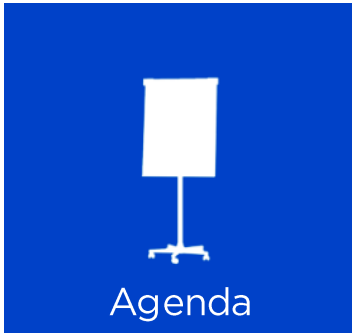
Slide Download



https://www.4tecture.ch/events/ddc25-net10

Agenda

- Intro
- ASP.NET Core 10
- Aspire 13
- File-based Apps / C# 14
- Q&A





.NET 10 for cloud apps

Intro

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.NET ecosystem momentum

7+ Million

Monthly active users

290k+

Pull requests merged to .NET

68k+

Contributors to .NET

Top 5

Highest velocity OSS project

Top 5

Programming language on GitHub

Sources: https://github.blog/news-insights/octoverse/octoverse-2024/#the-most-popular-programming-languages, https://github.com/cncf/velocity/blob/main/reports/top_projects_by_activity.txt

.NET 10 and C# 14

- Industry leading performance
- Supercharged SKD and libraries
- LTS Release → 3 Years of Support

Modern features for

- Web
- Cloud
- Mobile
- Cross-platform

Visual Studio 2026

Cloud Apps Requirements

- Density
  - Low memory footprint
  - High throughput
- Observability
  - Integration in monitoring / observability systems
  - Signals: Logs, Metrics, Traces
- Stability & Ecosystem
  - Battle-tested framework
  - Efficient development based on libraries
  - Extensibility
- Security


.NET 10 for cloud apps
ASP.NET Core
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ASP.NET Core – complete web framework

- **Blazor**
  Rich interactive Web UI
- **Minimal APIs**
  Fast HTTP APIs
- **SignalR**
  Real-time services
- **gRPC**
  Backend RPC calls
- **Middleware**
  Built-in & custom
- **Kestrel**
  High performance HTTP server

ASP.NET Core focus areas for .NET 10

Security

Observability & Diagnostics

Performance Improvements

Address pain-points

Why Passkeys

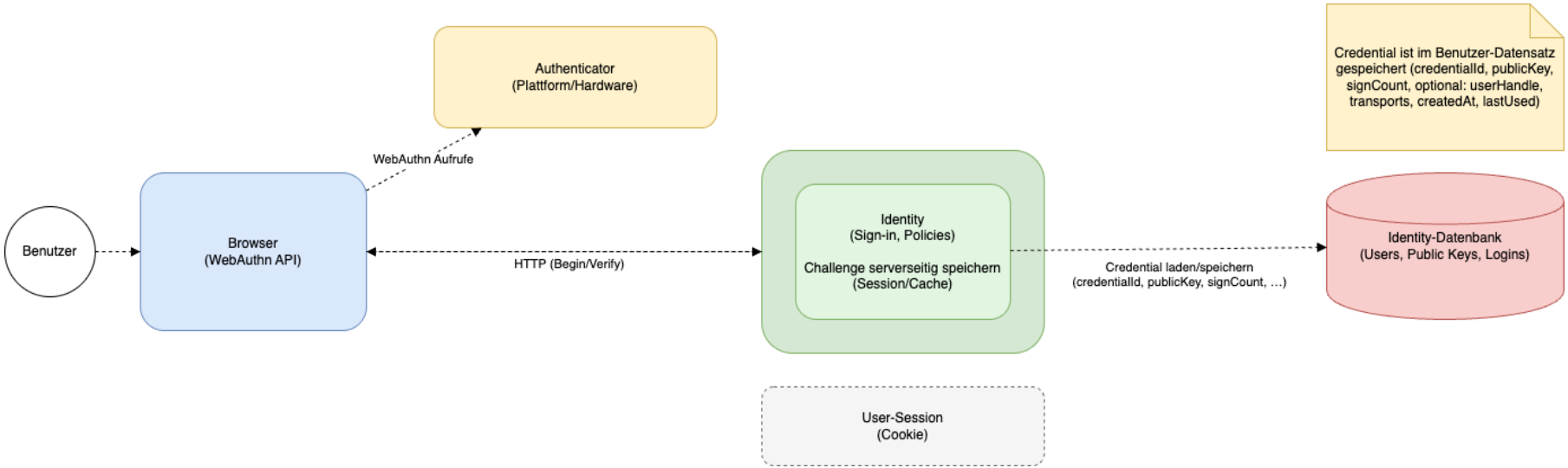
- Phishing-resistant
- Easy to use
- Secure by design

Passkeys are cryptographic credentials that replace traditional passwords

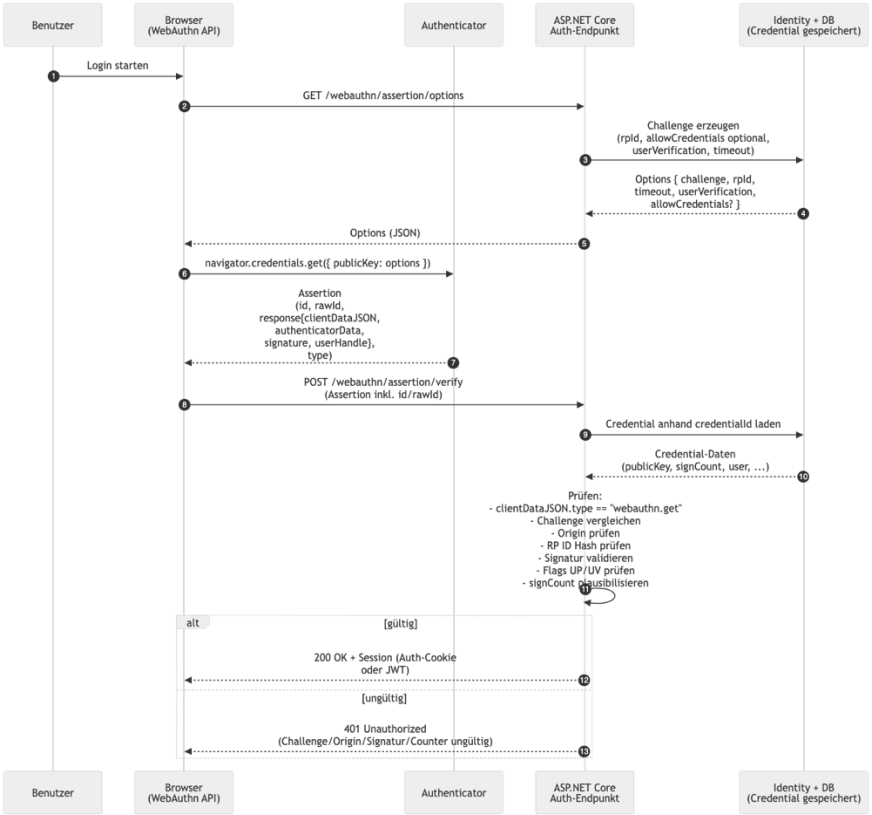
WebAuthN & Passkey Support

- Added Built-in support for Passkeys to ASP.NET Core Identity
  - Based on fido2-net-lib (OSS)
- Extend existing apps based on ASP.NET Core Identity to support passkeys

How Passkeys work...



Passkeys Flow




DEMO
Passkeys

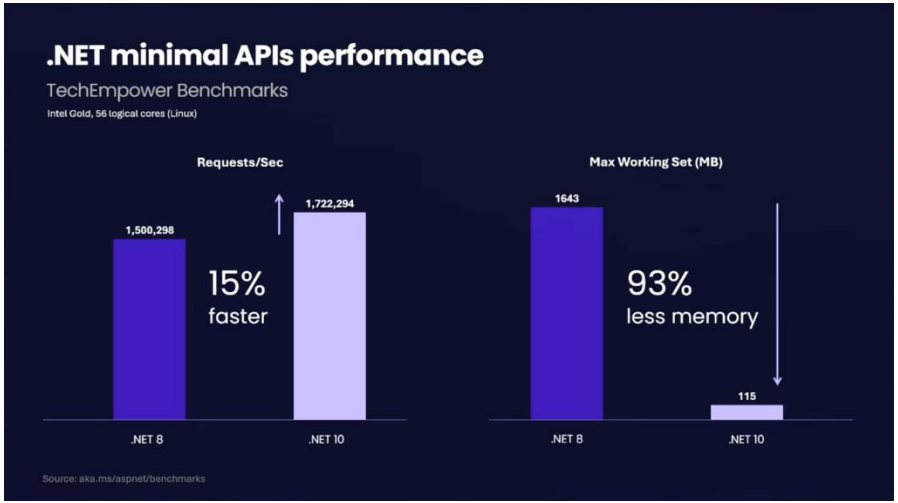
Observability

- Everything is based on Open Telemetry
- New metrics:
  - Microsoft.AspNetCore.Identity
  - Microsoft.AspNetCore.MemoryPool
- Improved Open Telemetry Support
- Aspire Dashboard with Service Defaults


DEMO
Observability

Performance

- General improvements in .NET
- Request throughput and memory usage
- Releasing memory from Kestrel memory pool
- API JSON deserialization performance



Better scaling, lower costs, smarter resource use

Sources: https://www.youtube.com/watch?v=LqUZB_rhyX4&list=PLdo4fOcmZ0oXtIlvq1tuORUtZqVG-HdCt&index=3, https://aka.ms/aspnet/benchmarks

Pain points & Gaps

- Minimal APIs
  - Minimal API Validation
  - Server-sent events
- OpenAPI Investments
  - Support for OpenAPI v3.1
  - XML doc comments-based OpenAPI descriptions
  - Support OpenAPI docs in JSON and YAML
- Various
  - JSON Patch with System.Text.Json
  - Generate correct unauthorized responses for APIs in Web apps (redirect or status code)


DEMO
Minimal APIs


DEMO
OpenAPI


DEMO
Server-Sent Events



.NET 10 for cloud apps

Aspire

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Challenges Aspires solves...

- Development Loop
  - Impossible to “clone and run”
  - No easy way to add new services
  - Local dev does not reflect the actual running system (production like env)
- Deployments
  - “Only in prod” observability
  - Changing assets
  - Complicated handoffs for architecture changes

Aspire is the tool for code-first, extensible, observable development and deployment

Aspirified App

Aspire AppHost + CLI

- stack defined in code
- typesafe, readable and debuggable
- Run locally and deploy it to real cloud services

Aspire Integrations

- Any code or service
- any executable, container, cloud resource or deployment pipeline

Aspire Dashboard

- See your entire stack running
- Logs, endpoints, environment variables
- Full Open Telemetry

Aspire

- From .NET Aspire to Aspire → Polyglot
- Aspire 13.0
- Single File App Host
- .Net, Javascript, and Python
- Flexible Connection Strings and certificate trust across languages and Containers

→ Opinionated defaults to get you started


DEMO
Aspire



.NET 10 for cloud apps

File-based Apps & C# 14

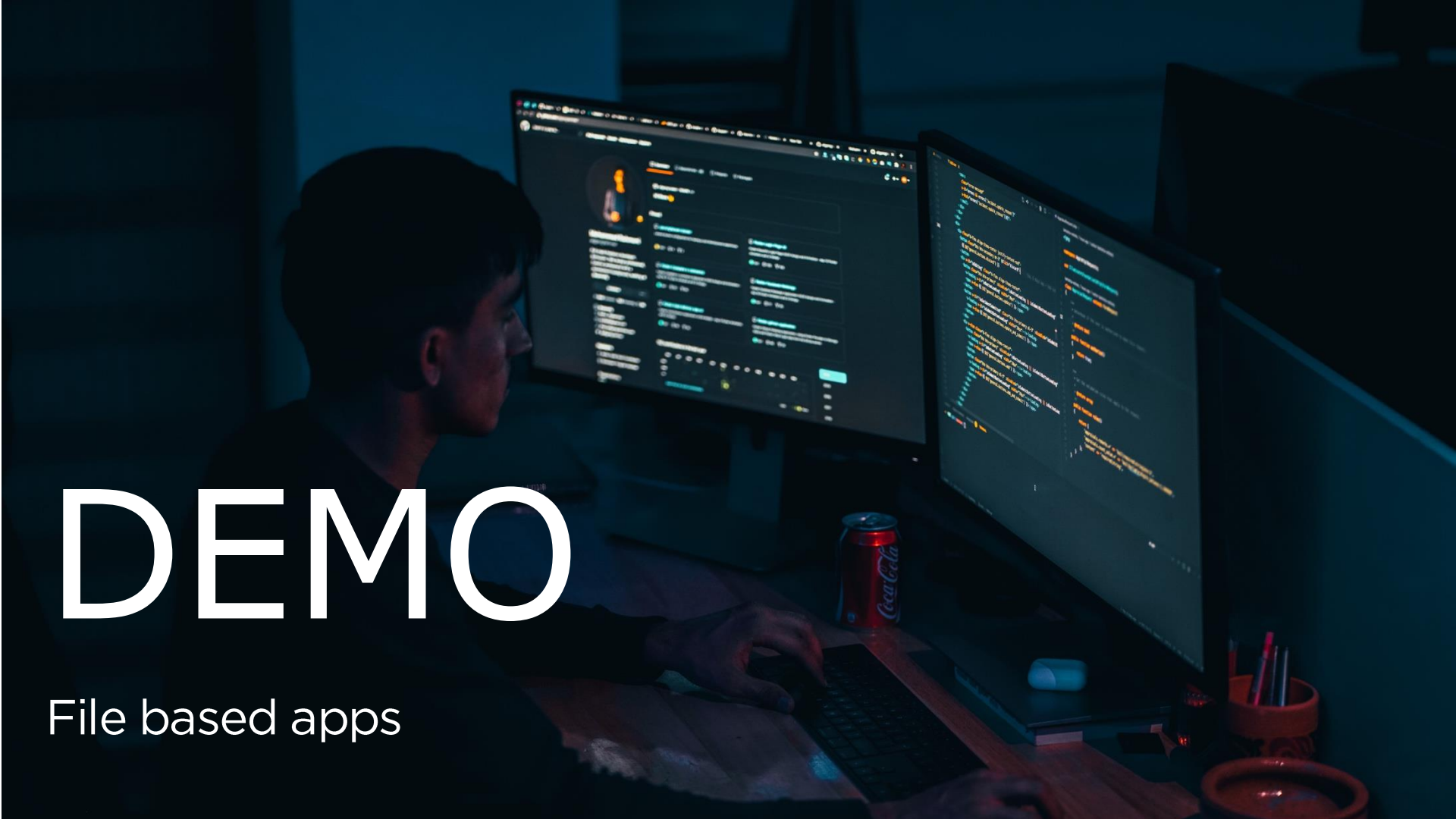
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File-based Apps

- Single file, no csproj
- NuGet package support
- shebang support

Usage:

- Automation
- Aspire AppHost
- ...



DEMO

File based apps

C# 14

- Null-conditional assignment
- Field backed properties
- Extension members
- Modifiers on simple lambda parameters
- Nameof supports unbound generic types
- Implicit conversions for Span<T> and ReadOnlySpan<T>
- Partial events and constructors
- User defined compound assignment operators


DEMO
C# 14



.NET 10 for cloud apps

Conclusion

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Conclusion

- Expected Performance Improvements
- Minimal APIs get long awaited features
- SSE simplifies push notifications
- Aspire gets mature, local dev and deployment with one tool / code-base
- File-based apps are interesting as script-replacement, tools and automations


.NET 10 for cloud apps
Q & A
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Thank you for your attention!

If you have any questions do not hesitate to contact us:

4tecture GmbH
Industriestrasse 25
CH-8604 Volketswil

Marc Müller
Principal Consultant

+41 44 508 37 00
info@4tecture.ch
www.4tecture.ch

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ProAuth

ReaFx


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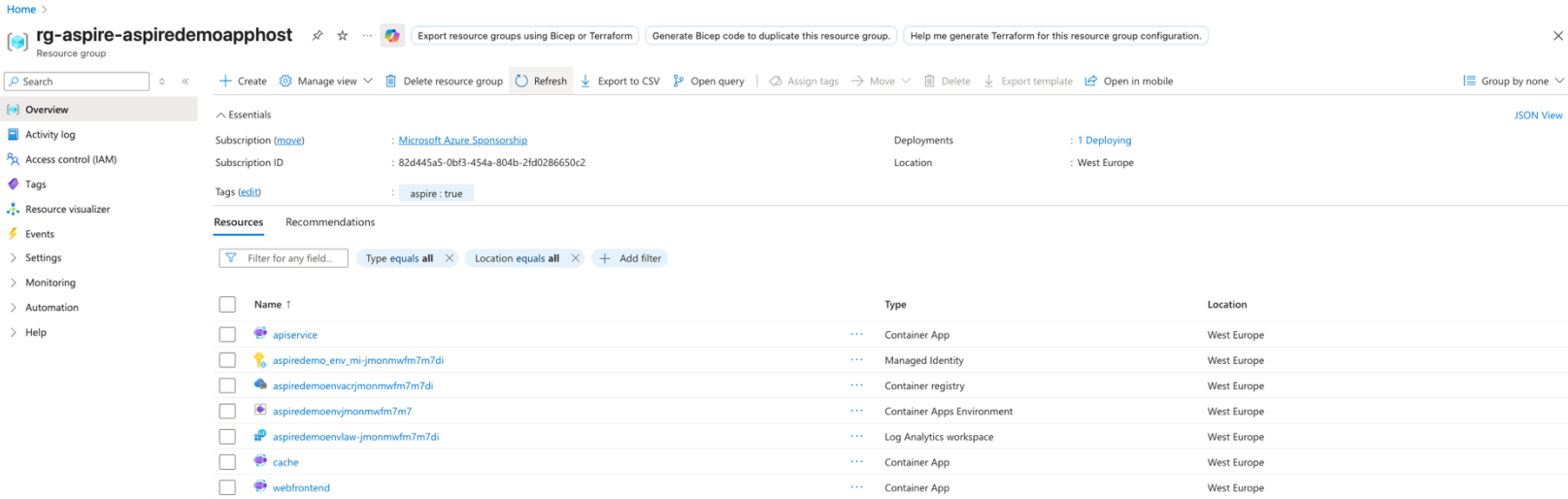

DEMO

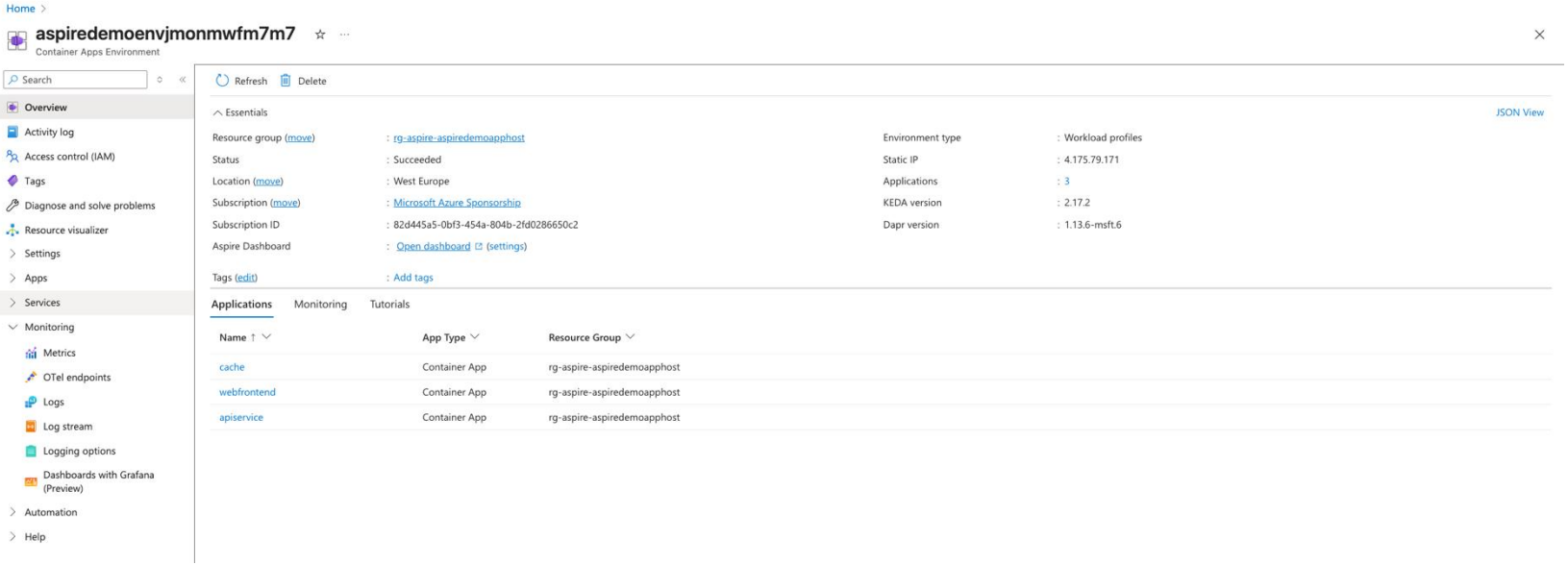
```
(base) marc@4tnb007 AspireDemo % aspire deploy
  12:56:19 (pipeline execution) → Starting pipeline execution...
  12:56:19 (deploy-prereq) → Starting deploy-prereq...
  12:56:19 (build-prereq) → Starting build-prereq...
  12:56:19 (build-prereq) ✓ build-prereq completed successfully
  12:56:19 (deploy-prereq) i [INF] Initializing deployment for environment 'Production'
  12:56:19 (deploy-prereq) i [INF] Parameter values saved to deployment state.
  12:56:19 (deploy-prereq) i [INF] Setting default deploy tag 'aspire-deploy-20251121115619' for compute resource(s).
  12:56:19 (deploy-prereq) ✓ deploy-prereq completed successfully
  12:56:19 (build-webfrontend) → Starting build-webfrontend...
  12:56:19 (build-apiservice) → Starting build-apiservice...
  12:56:19 (validate-azure-login) → Starting validate-azure-login...
  12:56:19 (build-webfrontend) i [INF] Building container image for resource webfrontend
  12:56:19 (build-apiservice) i [INF] Building container image for resource apiservice
  12:56:19 (build-webfrontend) i [INF] Building image: webfrontend
  12:56:20 (validate-azure-login) ✓ Azure CLI authentication validated successfully
  12:56:20 (create-provisioning-context) → Starting create-provisioning-context...
  12:56:20 (fetch-tenant) → Starting fetch-tenant...
  12:56:20 (fetch-tenant) → Fetching available tenants
  12:56:21 (fetch-tenant) ✓ Fetching available tenants (1.2s)
  12:56:21 (fetch-tenant) ✓ Found 6 available tenant(s)
Select your Azure tenant:
Tenant ID:  4tecture Demo Environment (mmueller4tecture.onmicrosoft.com) — eae84aff-29e1-48bc-8daa-57ecbca66884
  12:56:36 (build-webfrontend) i [INF] Building image for webfrontend completed
  12:56:36 (build-apiservice) i [INF] Building image: apiservice
  12:56:36 (build-webfrontend) ✓ build-webfrontend completed successfully
  12:56:36 (build-apiservice) i [INF] Building image for apiservice completed
  12:56:36 (build-apiservice) ✓ build-apiservice completed successfully
  12:56:36 (fetch-subscription) → Starting fetch-subscription...
  12:56:36 (fetch-subscription) → Fetching available subscriptions
  12:56:37 (fetch-subscription) ✓ Fetching available subscriptions (0.4s)
  12:56:37 (fetch-subscription) ✓ Found 2 available subscription(s)
Select your Azure subscription:
Subscription ID:  Microsoft Azure Sponsorship (82d445a5-0bf3-454a-804b-2fd0286650c2)
  12:56:40 (fetch-regions) → Starting fetch-regions...
  12:56:40 (fetch-regions) → Fetching supported regions
  12:56:42 (fetch-regions) ✓ Fetching supported regions (2.2s)
  12:56:42 (fetch-regions) ✓ Found 101 available region(s)
Select your Azure location and specify resource group:
Location:  West Europe
Resource group: (rg-aspire-aspiredemoapphost): rg-aspire-aspiredemoapphost
  12:57:08 (create-provisioning-context) i [INF] Using AzureCliCredential for provisioning.
  12:57:10 (create-provisioning-context) i [INF] Default subscription: Microsoft Azure Sponsorship (/subscriptions/82d445a5-0bf3-454a-804b-2fd0286650c2)
  12:57:10 (create-provisioning-context) i [INF] Tenant: eae84aff-29e1-48bc-8daa-57ecbca66884
  12:57:10 (create-provisioning-context) i [INF] Creating resource group rg-aspire-aspiredemoapphost in westeurope...
  12:57:11 (create-provisioning-context) i [INF] Resource group rg-aspire-aspiredemoapphost created.
  12:57:11 (create-provisioning-context) ✓ create-provisioning-context completed successfully
  12:57:11 (provision-aspiredemo-env) → Starting provision-aspiredemo-env...
  12:57:11 (provision-aspiredemo-env) → Deploying aspiredemo-env
  13:02:39 (provision-aspiredemo-env) ✓ Successfully provisioned aspiredemo-env (327.2s)
  13:02:39 (provision-aspiredemo-env) ✓ provision-aspiredemo-env completed successfully
  13:02:39 (provision-cache-containerapp) → Starting provision-cache-containerapp...
  13:02:39 (provision-cache-containerapp) → Deploying cache-containerapp
  13:02:39 (login-to-acr-aspiredemo-env) → Starting login-to-acr-aspiredemo-env...
  13:02:39 (login-to-acr-aspiredemo-env) → Logging in to aspiredemoenvacrjmonmwfm7m7di
  13:02:42 (login-to-acr-aspiredemo-env) ✓ Successfully logged in to aspiredemoenvacrjmonmwfm7m7di (3.0s)
  13:02:42 (login-to-acr-aspiredemo-env) ✓ login-to-acr-aspiredemo-env completed successfully
  13:02:42 (push-webfrontend) → Starting push-webfrontend...
  13:02:42 (push-apiservice) → Starting push-apiservice...
  13:02:42 (push-apiservice) → Pushing apiservice to aspiredemoenvacrjmonmwfm7m7di
  13:02:42 (push-webfrontend) → Pushing webfrontend to aspiredemoenvacrjmonmwfm7m7di
  13:02:42 (push-apiservice) i [INF] Docker tag for apiservice -> aspiredemoenvacrjmonmwfm7m7di.azurecr.io/apiservice:aspire-deploy-20251121115619 succeeded.
  13:02:42 (push-webfrontend) i [INF] Docker tag for webfrontend -> aspiredemoenvacrjmonmwfm7m7di.azurecr.io/webfrontend:aspire-deploy-20251121115619 succeeded.
  13:03:00 (provision-cache-containerapp) ✓ Successfully provisioned cache-containerapp (21.2s)
  13:03:00 (provision-cache-containerapp) ✓ provision-cache-containerapp completed successfully
  13:03:00 (print-cache-summary) → Starting print-cache-summary...
  13:03:00 (print-cache-summary) i [INF] Successfully deployed cache to Azure Container Apps environment aspiredemo-env. No public endpoints were configured.
  13:03:00 (print-cache-summary) ✓ print-cache-summary completed successfully
  13:03:29 (push-apiservice) i [INF] Docker push for aspiredemoenvacrjmonmwfm7m7di.azurecr.io/apiservice:aspire-deploy-20251121115619 succeeded.
  13:03:29 (push-apiservice) ✓ Successfully pushed apiservice to aspiredemoenvacrjmonmwfm7m7di.azurecr.io/apiservice:aspire-deploy-20251121115619 (47.0s)
  13:03:29 (push-apiservice) ✓ push-apiservice completed successfully
  13:03:29 (provision-apiservice-containerapp) → Starting provision-apiservice-containerapp...
  13:03:29 (provision-apiservice-containerapp) → Deploying apiservice-containerapp
  13:03:30 (push-webfrontend) i [INF] Docker push for aspiredemoenvacrjmonmwfm7m7di.azurecr.io/webfrontend:aspire-deploy-20251121115619 succeeded.
  13:03:30 (push-webfrontend) ✓ Successfully pushed webfrontend to aspiredemoenvacrjmonmwfm7m7di.azurecr.io/webfrontend:aspire-deploy-20251121115619 (48.7s)
  13:03:30 (push-webfrontend) ✓ push-webfrontend completed successfully
  13:03:30 (provision-webfrontend-containerapp) → Starting provision-webfrontend-containerapp...
  13:03:30 (provision-webfrontend-containerapp) → Deploying webfrontend-containerapp
  13:03:50 (provision-apiservice-containerapp) ✓ Successfully provisioned apiservice-containerapp (21.4s)
  13:03:50 (provision-apiservice-containerapp) ✓ provision-apiservice-containerapp completed successfully
  13:03:50 (print-apiservice-summary) → Starting print-apiservice-summary...
  13:03:50 (print-apiservice-summary) i [INF] Successfully deployed apiservice to Azure Container Apps environment aspiredemo-env. No public endpoints were configured.
  13:03:50 (print-apiservice-summary) ✓ print-apiservice-summary completed successfully
  13:03:51 (provision-webfrontend-containerapp) ✓ Successfully provisioned webfrontend-containerapp (20.4s)
  13:03:51 (provision-webfrontend-containerapp) ✓ provision-webfrontend-containerapp completed successfully
  13:03:51 (provision-azure-bicep-resources) → Starting provision-azure-bicep-resources...
  13:03:51 (print-webfrontend-summary) → Starting print-webfrontend-summary...
  13:03:51 (print-webfrontend-summary) i [INF] Successfully deployed webfrontend to https://webfrontend.gentlepebble-1766b00c.westeurope.azurecontainerapps.io
  13:03:51 (print-webfrontend-summary) ✓ print-webfrontend-summary completed successfully
  13:03:51 (provision-azure-bicep-resources) ✓ provision-azure-bicep-resources completed successfully
  13:03:51 (print-dashboard-url-aspiredemo-env) → Starting print-dashboard-url-aspiredemo-env...
  13:03:51 (print-dashboard-url-aspiredemo-env) ✓ Dashboard available at dashboard_URL
  13:03:51 (deploy) → Starting deploy...
  13:03:51 (deploy) ✓ deploy completed successfully
  13:03:51 (pipeline execution) ✓ Completed successfully
──────────────────────────────────────────────────
✓ 23/23 steps succeeded • Total time: 451.6s
```

```
✓ 23/23 steps succeeded • Total time: 451.6s

Steps Summary:
  451.6 s  ✓ pipeline execution
  327.2 s  ✓ provision-aspiredemo-env
   51.3 s  ✓ create-provisioning-context
   48.7 s  ✓ push-webfrontend
   47.0 s  ✓ push-apiservice
   21.4 s  ✓ provision-apiservice-containerapp
   21.2 s  ✓ provision-cache-containerapp
   20.4 s  ✓ provision-webfrontend-containerapp
   17.1 s  ✓ build-apiservice
   17.1 s  ✓ build-webfrontend
    3.0 s  ✓ login-to-acr-aspiredemo-env
    2.2 s  ✓ fetch-regions
    1.2 s  ✓ fetch-tenant
    0.9 s  ✓ validate-azure-login
    0.4 s  ✓ fetch-subscription
    0.0 s  ✓ deploy-prereq
    0.0 s  ✓ provision-azure-bicep-resources
    0.0 s  ✓ print-webfrontend-summary
    0.0 s  ✓ print-cache-summary
    0.0 s  ✓ print-apiservice-summary
    0.0 s  ✓ build-prereq
    0.0 s  ✓ print-dashboard-url-aspiredemo-env
    0.0 s  ✓ deploy

✓ PIPELINE SUCCEEDED
```


Home
rg-aspire-aspiredemoapphost
Resource group
Export resource groups using Bicep or Terraform
Generate Bicep code to duplicate this resource group.
Help me generate Terraform for this resource group configuration.
Search
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Monitoring
Automation
Help
Create
Manage view
Delete resource group
Refresh
Export to CSV
Open query
Assign tags
Move
Delete
Export template
Open in mobile
Group by none
Essentials
JSON View
Subscription (move) : Microsoft Azure Sponsorship
Subscription ID : 82d445a5-0bf3-454a-804b-2fd0286650c2
Tags (edit) : aspire : true
Deployments : 1 Deploying
Location : West Europe
Resources
Recommendations
Filter for any field...
Type equals all
Location equals all
Add filter
Name
Type
Location
apiservice
Container App
West Europe
aspiredemo_env_mi-jmonmwfm7m7di
Managed Identity
West Europe
aspiredemoenvacrjmonmwfm7m7di
Container registry
West Europe
aspiredemoenvjmonmwfm7m7
Container Apps Environment
West Europe
aspiredemoenvlaw-jmonmwfm7m7di
Log Analytics workspace
West Europe
cache
Container App
West Europe
webfrontend
Container App
West Europe


Home
aspiredemoenvjmonmwfm7m7
Container Apps Environment
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Apps
Services
Monitoring
Metrics
OTel endpoints
Logs
Log stream
Logging options
Dashboards with Grafana (Preview)
Automation
Help
Refresh
Delete
Essentials
JSON View
Resource group (move) : rg-aspire-aspiredemoapphost
Status : Succeeded
Location (move) : West Europe
Subscription (move) : Microsoft Azure Sponsorship
Subscription ID : 82d445a5-0bf3-454a-804b-2fd0286650c2
Aspire Dashboard : Open dashboard (settings)
Environment type : Workload profiles
Static IP : 4.175.79.171
Applications : 3
KEDA version : 2.17.2
Dapr version : 1.13.6-msft.6
Tags (edit) : Add tags
Applications
Monitoring
Tutorials
Name
App Type
Resource Group
cache
Container App
rg-aspire-aspiredemoapphost
webfrontend
Container App
rg-aspire-aspiredemoapphost
apiservice
Container App
rg-aspire-aspiredemoapphost